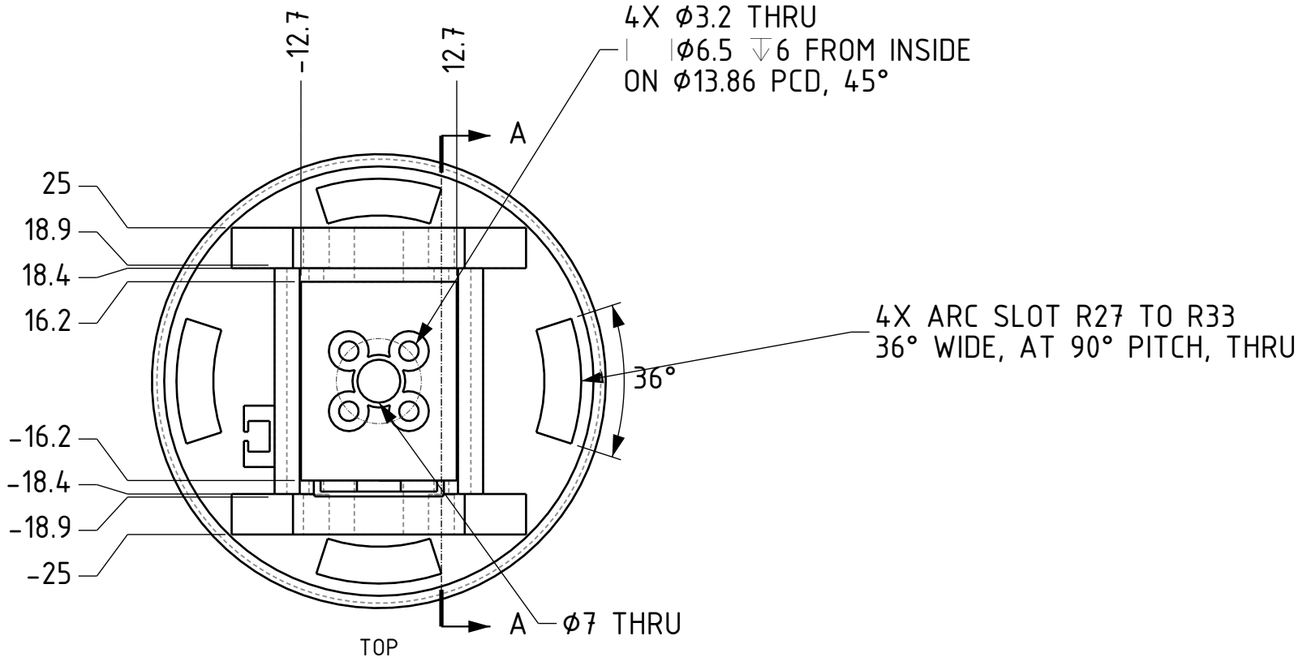
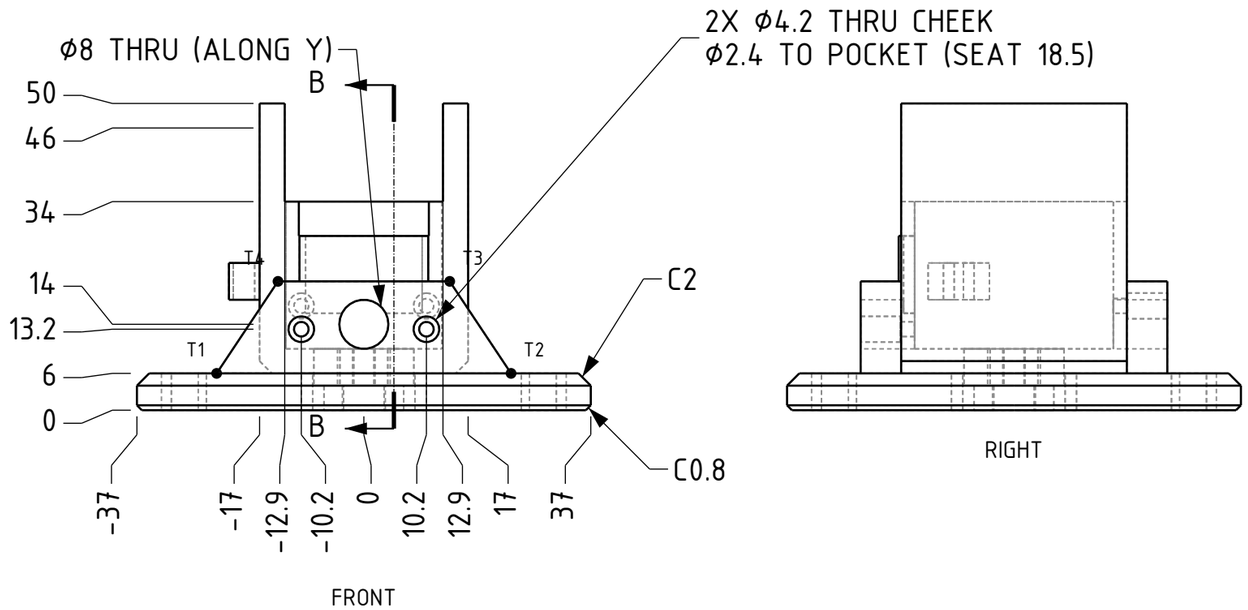




CHEEK PROFILE (X, Z)

PT	X	Z
T1	-24	6
T2	24	6
T3	14	21
T4	-14	21

FEATURES (in order)

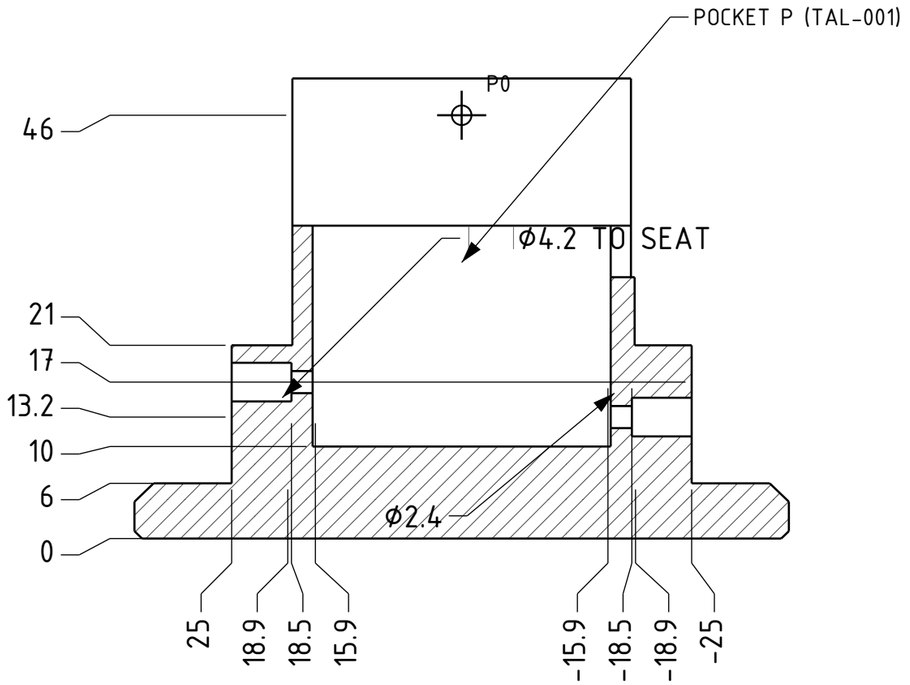
1. Turntable Ø74, z 0 to 6; C2 top edge, C0.8 bottom edge.
2. Cradle: box x ±17, y ±18.4, z 5 to 34.
3. Posts: boxes x -17 to -12.9 and 12.9 to 17, y ±18.4, z 30 to 50.
4. Cheeks: profile T1 to T4, y 18.4 to 25 and -25 to -18.4 (fused to the cradle walls).
5. Doubler on the -Y cradle wall: box x ±10.5, y -18.8 to -18.4, z 14.8 to 34 (keeps the wall beside the back pad channel 0.8 thick).
6. Cable clip (TAL-002) on the -X face: centre (-17, -9, 21), cable along Z.
7. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 46): servo +Z (shaft) along +Y, servo +X along +Z (body hangs down), servo +Y along +X.
8. Servo opening: cut box x ±12.7, y ±16.2, from z 10 up. Servo entry channels in the -Y cradle wall: back pad channel x ±9.5, y -18 to -16.2, from z 15.8 up; plug slot through the wall x ±10.6, from z 28.4 up.
9. Cable hole Ø8 along Y, through all, at x 0, z 14.
10. Pan horn: Ø7 centre through. 4X Ø3.2 through on Ø13.86 PCD at 45°, counterbore Ø6.5 from the floor (z 10) down to z 4 (M3 × 6 from inside the cradle); centre opened to Ø8.6 over the same depth (no thin web between the counterbores and the centre hole).
11. Turntable windows: 4X arc slot R27 to R33, ±18° about 0°, 90°, 180°, 270°, through.
12. Case screw holes (TAL-002, seat 18.5) through the cheeks: +Y cheek at x ±10.2, z 17; -Y cheek at x ±10.2, z 13.2.

NOTES

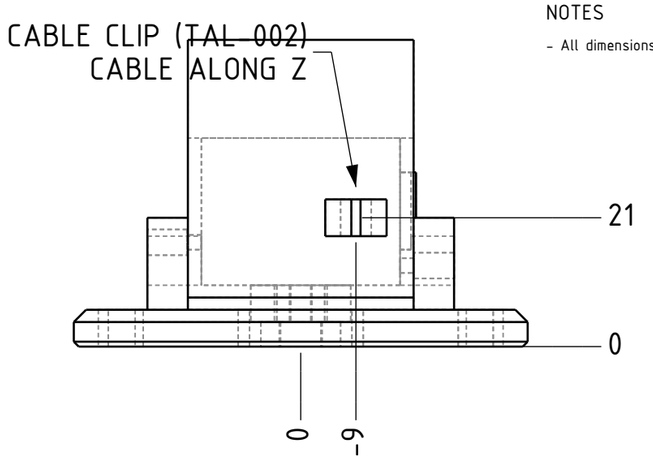
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the shoulder servo from the top, bottom end first with the shaft to +Y, before its horns and plugs; then 4X M2 × 6 case screws through the cheeks.
- Print: turntable down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 37 g
PART Shoulder (turntable + cradle) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-07
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 8 of 21



SECTION A-A



LEFT SCALE 1:1

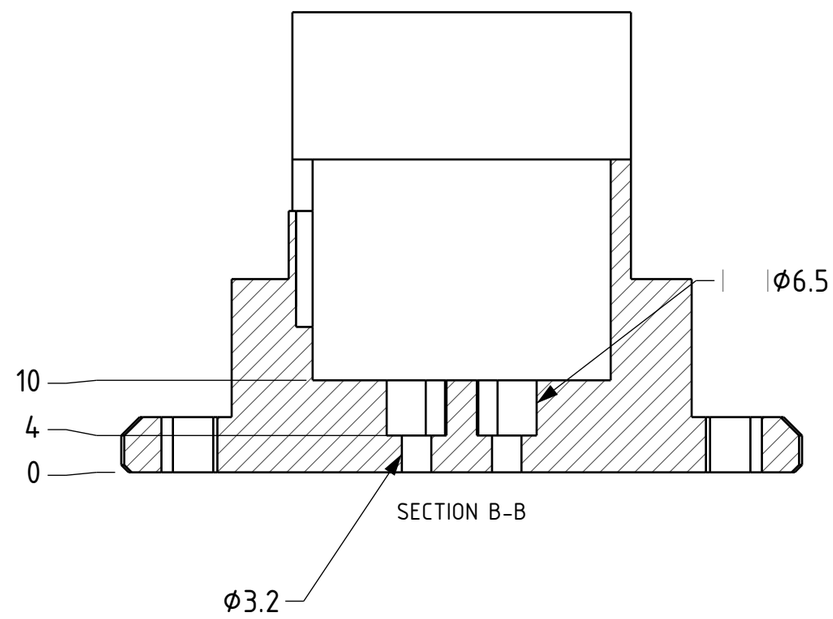
SECTIONS

A-A: plane x 10.2 through one hole of each case screw row (cutting line on sheet 1, top view), looking toward +X. The other holes mirror at x -10.2.

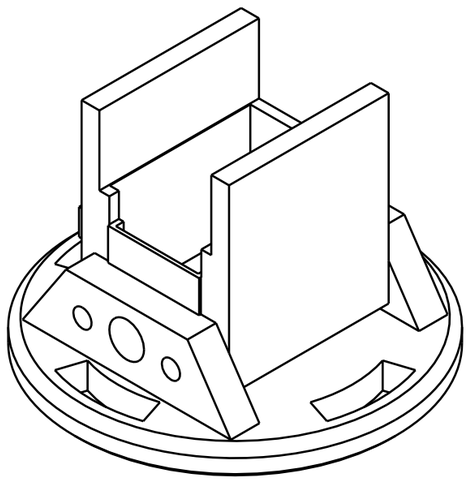
B-B: plane x 4.9 through two pan horn screw holes (cutting line on sheet 1, front view).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



SECTION B-B



ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Shoulder (turntable + cradle) (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-07
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 9 of 21